

Supplement: variational aggregation objective and weight control

## Supplement: variational aggregation objective and weight control I

This supplement gives the full derivation behind sec. 5.8.2: the server-side aggregator `robust_aggregate` is a heuristic, and a single change of divergence direction turns it into block-coordinate descent on a stated free energy with a derived, redescending effective-weight update. We work throughout with categorical local posteriors  $q_n(s)$  over the shared latent factor, base weights  $w_n > 0$ , robustness  $c > 0$ , and the consensus  $q$  on the probability simplex. Let  $\lambda > 0$  be the entropy-weight coefficient, with the current default at  $\lambda = 1.0$ .

## Why the sharp heuristic is not yet variational I

The sharp server rule is empirically strong, but this repository has not established a variational certificate for it. The heuristic of eq. 7 alternates a *reverse-KL* weight update

$a_n \leftarrow w_n \exp(-c \text{KL}(q_n \parallel q))$  with the log-linear consensus  $q \leftarrow \text{softmax}(\sum_n a_n \log q_n)$ . For the natural direct objective  $\sum_n a_n \text{KL}(q_n \parallel q) + \frac{1}{c} \text{KL}_{\text{gen}}(a \parallel w)$ , the reverse-KL rule is the  $a$ -minimizer, but the  $q$ -minimizer is the *arithmetic* (linear) pool  $q \propto \sum_n a_n q_n$ , not the log-linear pool. The executable orientation witness confirms this finite-simplex mismatch. The following proposition goes further, while retaining a deliberately narrow scope.

## Why the sharp heuristic is not yet variational I

$$Q(a; s) = \text{softmax}\left(\sum_n a_n \log s_n\right) \quad (1)$$

The proposed raw  $q$ -block is the map in eq. 1.

$$F(q, a; s, w) = \sum_n a_n \text{KL}(q \| s_n) + R(a, w) + G(q). \quad (2)$$

### Proposition (Scoped separable raw-log-pool no-go)

*For categorical state dimension  $K \geq 2$ , no objective of the declared displayed form with  $G$  continuously differentiable,  $R$  independent of  $q$  and the  $s_n$ , and  $G$  independent of  $a$  and the  $s_n$ , has  $Q(a; s)$  as its  $q$ -coordinate minimizer for every interior raw  $a \in \mathbb{R}_{>0}^N$  and every interior local-posterior collection  $s$ .*

*Consequently, this objective class cannot realize both block maps of the implemented raw-weight heuristic.*

## Why the sharp heuristic is not yet variational I

*Proof sketch.* Fix any non-uniform interior  $q$ , a positive scalar  $\alpha$ , and one local posterior constructed in eq. 3:

$$s_i^{(\alpha)} = \frac{q_i^{1/\alpha}}{\sum_j q_j^{1/\alpha}}. \quad (3)$$

Then  $Q(\alpha; s^{(\alpha)}) = q$ . Writing  $\Pi$  for projection onto the tangent space of the simplex, first-order stationarity of eq. 2 at that same  $q$  requires  $\Pi[(\alpha - 1) \log q + \nabla G(q)] = 0$ . The unit-scale construction forces  $\Pi \nabla G(q) = 0$ ; any different positive scale then forces  $\Pi \log q = 0$ , contradicting the non-uniform choice of  $q$ . The executable witness records both exact log-pool identities and the nonzero tangential contradiction.

## Why the sharp heuristic is not yet variational I

A companion witness also blocks the obvious normalized-weight escape within the same natural data-term class: two interior consensuses yield the same normalized reverse-KL weights but different forward-KL data-term differences, so a  $q$ -independent differentiable  $R(a, w)$  cannot satisfy both simplex-stationarity equations. The implementation itself uses raw effective weights, so this companion is a scope check rather than a description of the production update.

# Why the sharp heuristic is not yet variational I

Table 1: Formal MAJ-1 witness inventory. These are deterministic finite-simplex proof artifacts, not empirical estimates; no resampling interval or deployment claim is implied.

| Formal artifact             | Executable source                                  | Scope                                                                    |
|-----------------------------|----------------------------------------------------|--------------------------------------------------------------------------|
| Raw-log-pool contradiction  | <code>server_theory.py</code> : raw witness        | The declared raw $q$ -block map over every interior input                |
| Normalized-weight companion | <code>server_theory.py</code> : normalized witness | The normalized reparameterization of the same forward-KL data-term class |
| Typed source report         | report <code>formal_no_go</code> field             | Deterministic witness metadata, separate from the empirical attack grid  |

## Why the sharp heuristic is not yet variational I

Table tbl. 1 records the deterministic implementation surfaces that bind this scoped result to the typed analysis report.

The proposition does **not** say that no objective of any kind exists. It does not exclude nonseparable  $q$ - $a$  couplings, source-dependent terms, non-differentiable constructions, or objectives that encode selected fixed points without reproducing the update blocks for all interior inputs. Thus sec. 5.8 retains the heuristic label and claims only the recovery limit, the scoped negative result above, and conditional empirical behavior — never a bounded-influence property or an objective-backed status.

### Definition (Aggregation free energy)

For  $c > 0$ ,  $\lambda > 0$ , consensus  $q$ , and effective weights  $a = (a_n)$ ,  $a_n \geq 0$ , define  $F_\lambda(q, a)$  as in (8) with the forward cross-entropy  $\text{CE}(q, q_n) = -\sum_i q_i \log q_{n,i}$ , the consensus entropy  $H(q)$ , and the generalized KL  $\text{KL}_{\text{gen}}(a \parallel w) = \sum_n [a_n \log(a_n/w_n) - a_n + w_n]$ .

**The  $q$ -block.** For  $\lambda > 0$ , fixing  $a$ , the  $q$ -dependent part of  $F_\lambda$  is  $\sum_n a_n \text{CE}(q, q_n) - \lambda H(q) = \sum_i q_i [\lambda \log q_i - \sum_n a_n \log q_{n,i}]$ . Adding a Lagrange multiplier for  $\sum_i q_i = 1$  and differentiating gives  $\lambda \log q_i + \lambda - \sum_n a_n \log q_{n,i} + \mu = 0$ , i.e.

## Aggregation free energy and its block minimizers I

$$q_i \propto \exp\left(\frac{1}{\lambda} \sum_n a_n \log q_{n,i}\right) = \text{softmax}\left(\frac{1}{\lambda} \sum_n a_n \log q_n\right)_i, \quad (4)$$

the product of the weighted experts — the consensus update of eq. 9. At the default  $\lambda = 1.0$ , the  $-H(q)$  term sharpens the weighted geometric mean into the product-of-experts (the entropy bonus that makes the project's log-linear pool a product rather than a geometric average).

**The  $a$ -block.** Fixing  $q$ ,

$\partial F / \partial a_n = \text{CE}(q, q_n) + \frac{1}{c} \log(a_n / w_n) = 0$ , so

$$a_n = w_n \exp(-c \text{CE}(q, q_n)), \quad (5)$$

the weight update of eq. 9. Because

$\text{CE}(q, q_n) = H(q) + \text{KL}(q \parallel q_n)$ , the forward direction  $\text{KL}(q \parallel q_n)$  — not the heuristic's reverse  $\text{KL}(q_n \parallel q)$  — is the one consistent with the consensus update.

Each block update is the *exact* minimizer of its block, so alternating them is block-coordinate descent:  $F$  is non-increasing at every half-step. When the iterates converge, their fixed point is coordinatewise stationary.

## Aggregation free energy and its block minimizers I

The implementation keeps numerical failure handling outside that theorem. If finite-precision underflow collapses all effective weights, it records a fallback event, substitutes the declared base weights to return a valid probability vector, and does not certify the substituted trajectory as converged. Such a trace is diagnostic evidence about the solver boundary, not an instance of the exact block-descent result.

### Theorem (Variational aggregation: descent, recovery, and effective-weight bound)

*Let  $c > 0$  and  $\lambda > 0$ . Each alternating update in (4)–(5) never increases  $F$ . Any converged fixed point is coordinatewise stationary. As  $c \rightarrow 0$  the generalized-KL penalty forces  $a_n \rightarrow w_n$  and the consensus is the tempered log-linear pool; at the default  $\lambda = 1.0$  it is the log-linear pool (6) exactly, so the variational aggregator shares the project log-linear-pool corner of (7). Under the qualified bridge of Section 5.8, this is only the categorical message-combination specialization, not the complete source protocol. Finally, the effective weights satisfy  $a_n = w_n \exp(-c \text{CE}(q, q_n)) \leq w_n$  with  $a_n \rightarrow 0$  as  $\text{KL}(q \parallel q_n) \rightarrow \infty$ . Thus the raw effective-weight update is bounded and redescending relative to the realized consensus. This statement does not by itself establish a bounded influence function or finite gross-error sensitivity for the normalized consensus estimator.*

## Formal properties of the conservative server rule I

The objective  $F$  is biconvex (each block convex, the coupling  $\sum_n a_n \text{CE}(q, q_n)$  bilinear), so the result concerns monotone block updates and converged coordinatewise fixed points, not guaranteed convergence to or certification of a global minimum.

### **The effective-weight regime, and why multi-start matters.**

The weight bound  $a_n \leq w_n$  is unconditional ( $\text{CE}(q, q_n) \geq 0$  always). The *collapse*  $a_n \rightarrow 0$  is driven by the agent's divergence *from the realized consensus*  $\text{KL}(q \parallel q_n)$ , and the consensus itself depends on the weights. Because  $F$  is biconvex, this couples into a subtlety an adversarial review of this work surfaced: a *near-one-hot* saboteur (contamination rate  $\rightarrow 1$ ) already captures the product-of-experts, so a descent seeded *at* that pool stays in a consensus-capture basin (high  $F$ ) where the saboteur keeps its weight — even against

## Formal properties of the conservative server rule I

an honest majority. The repair is to search the stated objective more carefully: `variational_aggregate` runs **multi-start** block-coordinate descent (the pool, the uniform belief, and the arithmetic-mean seeds) and returns the lowest-observed- $F$  converged candidate. In the configured colony, the uniform/arithmetic seeds reach a lower- $F$  *vetoing* basin, so the saboteur is suppressed even at the simplex vertex (pinned qualitatively by the near-vertex multi-start test) — the  $267.1\times$  suppression of fig. 15 is measured across the swept contamination grid, whose most extreme point sits just below rate 1. What remains fundamental to every robust fusion rule, and is not claimed away: with no honest majority — a colony split with no anchoring plurality — there is no truth to recover. The observed suppression is conditional on the tested colonies and the fixed point

selected by a finite multi-start heuristic.

fig. 1 makes the capture and the escape concrete on a near-vertex colony: the single (log-linear-pool) start settles at  $F = 1.3092$  (the capture basin, where the saboteur keeps its weight), while the multi-start descent reaches the genuinely lower  $F = -0.2305$  vetoing basin — a gap of 1.5397 nats that is exactly the difference between trusting the natural seed and solving the stated objective properly.

# Formal properties of the conservative server rule I

**Multi-start escapes the near-vertex capture basin**

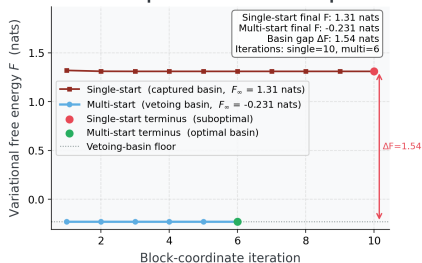


Figure 1: Variational free-energy descent on a near-vertex adversarial colony. Source relation: original project objective-descent diagnostic; estimand: free energy  $F$  in nats by iteration; uncertainty: none for deterministic seeded runs. The figure compares the single (log-linear-pool) start versus the multi-start descent. The x-axis is the block-coordinate iteration; the y-axis is the free energy  $F$  in nats. The single-start trajectory settles in the high- $F$  capture basin ( $F = 1.3092$ , the saboteur retains weight); the multi-start trajectory reaches the lower- $F$  vetoing basin ( $F = -0.2305$ ), a gap of 1.5397 nats. Deterministic seeded runs, so no error band.

## Numerical witnesses for descent and influence bounds I

The analysis pipeline runs `variational_aggregate` at robustness  $c = 1.50$  on a contaminated colony and records the free energy after each iteration. The descent falls from  $F = 3.2458$  to  $F = 2.3780$  (a monotone drop of 0.8678 over 11 iterations, converged: Yes); the largest single-step *increase* is  $8.88 \times 10^{-16}$ , machine zero — the monotonicity of the theorem, witnessed numerically and drawn in fig. 14.

## Numerical witnesses for descent and influence bounds I

For the effective-weight diagnostic, one agent is drifted from healthy toward a confident-wrong delta and its normalized influence is read at each drift. Clean, it carries 0.143 of the pool; at the most extreme swept drift it carries below 0.001 — a factor of 267.1 (computed from the unrounded influences, not the display-rounded values above) below the fixed 0.143 the naive pool would still grant it (fig. 15). This makes the redescending normalized-weight behavior visible on the tested path; it is not an estimator-level B-robustness proof.

## Tempered aggregation family for the accuracy-guarantee trade I

The aggregator of `sec. sec:supp-derivation` fixes the entropy term at unit weight. Relaxing that single coefficient generates a one-parameter *tempered* family. Introduce an entropy weight  $\lambda > 0$  — the **inverse temperature is**  $1/\lambda$  — and minimize

$$F_\lambda(q, a) = \sum_n a_n \text{CE}(q, q_n) - \lambda H(q) + \frac{1}{c} \text{KL}_{\text{gen}}(a \| w). \quad (6)$$

## Tempered aggregation family for the accuracy-guarantee trade I

Repeating the  $q$ -block derivation of eq. 4 with the entropy scaled by  $\lambda$  leaves the  $a$ -block **untouched** and tempers only the consensus update:

$$q \propto \exp\left(\frac{1}{\lambda} \sum_n a_n \log q_n\right), \quad a_n = w_n \exp(-c \text{CE}(q, q_n)). \quad (7)$$

The  $\lambda \downarrow 0$  endpoint is separately implemented as a deterministic tied-argmax rule; it is not obtained by substituting  $\lambda = 0$  into eq. 6 or eq. 7.

## Tempered aggregation family for the accuracy-guarantee trade I

The weight update is **independent of**  $\lambda$ : the bound  $a_n \leq w_n$  with collapse  $a_n \rightarrow 0$  as  $\text{KL}(q \parallel q_n) \rightarrow \infty$  is unchanged, so the **raw effective-weight bound of sec. sec:supp-theorem holds for every**  $\lambda > 0$ . At  $\lambda = 1.0$  the temperature is unity and eq. 7 is **identical** to the current axis-3 aggregator eq. 9 — the default is bit-identical, not merely close. The  $c \rightarrow 0$  recovery of sec. sec:supp-theorem generalizes to the *tempered* log-linear pool  $q \propto \exp(\frac{1}{\lambda} \sum_n w_n \log q_n)$ ; at  $\lambda = 1.0$  this is exactly  $\text{softmax}(\sum_n w_n \log q_n)$  — the project's **log-linear pool** eq. 6. Under the shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 5.8, that pool is a categorical specialization of Friston Eq. 7's message-combination term, not a

## Tempered aggregation family for the accuracy-guarantee trade I

reconstruction of the complete source protocol.

Positive-temperature members away from that default are tempered pools, not Friston Eq. 7 itself; the project recovery checks, including ISC-10, remain project-local.

A small empirical sweep over  $\lambda \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1\}$  on 10 contaminated colonies (5 agents, 2 adversarial) asks whether a single  $\lambda^*$  makes the conservative aggregator narrow the gap to the sharp `robust_aggregate` point-accuracy. The closest observed weight is  $\lambda^* = 0.3$  with an accuracy gap of 0.0008. **A  $\lambda^*$  narrows the tested accuracy gap while preserving the stated weight bound on this grid.** If no  $\lambda$  closes that gap while preserving the derived weight update, the result is the conservatism trade-off of sec. 8.13, not a defect to hide.